

Codon Routing: Mapping the 64-Codon Genetic Code to 6D Jordan Algebra Channels

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September 2026

DOI: [10.5281/zenodo.22715355](https://doi.org/10.5281/zenodo.22715355)

Abstract

We present a computational mapping from the 64-codon genetic code to six-dimensional Jordan algebra routing channels. Each codon is encoded as a 6-bit base-4 integer (first base most significant), with chemistry-derived qubit coordinates computed from integer-scaled molecular weights. The routing system assigns each codon to one of six channels (E2, E6, E7, E5, E4, E0) based on amino acid properties, with cross-wiring to the octonion basis, 3-qubit quantum states, and a 15-layer central row structure. The Smith chart boundary uses $\Gamma(z) = (z - 1)/(z + 1)$ for impedance matching, while the self-inverse Möbius checksum uses $\Gamma(z) = (1 - z)/(1 + z)$ satisfying $\Gamma(\Gamma(z)) = z$. We verify the routing with 37 Zig tests, 6 Q# quantum witness operations, and 16 f64 sidecar validation tests. The mapping is a framework-internal heuristic construction, not an established biological mechanism.

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1 Introduction

The genetic code maps 64 three-letter codons (combinations of A, C, G, U/T) to 20 amino acids and stop signals [1]. This paper presents a computational framework that maps these 64 codons to six-dimensional Jordan algebra routing channels, connecting biological information processing to octonionic algebraic structures [2, 5]. The computational arithmetic uses Q128.128 fixed-point arithmetic [4], and the algebraic framework is described in a companion paper [3].

1.1 Scientific Scope

The codon routing system is a **heuristic mapping**—a framework-internal construction that assigns codons to algebraic channels based on amino acid properties. It is **not** an established biological mechanism. The mapping is verified computationally (37 Zig tests, 6 Q# witnesses, 16 f64 validations), but this verification confirms the internal consistency of the construction, not its biological validity.

2 Genetic Code Structure

2.1 The 64 Codons

The standard genetic code has 4 bases $\{A, C, G, U\}$ and 3-letter codons, giving $4^3 = 64$ codons:

$$\text{Codon} = (b_1, b_2, b_3), \quad b_i \in \{A, C, G, U\} \quad (1)$$

2.2 Base-4 Encoding

Each base is assigned a base-4 digit:

$$A = 0, \quad C = 1, \quad G = 2, \quad U/T = 3 \quad (2)$$

The 6-bit codon index (first base most significant):

$$\text{index} = b_1 \times 16 + b_2 \times 4 + b_3 \quad (3)$$

This gives indices 0 (AAA) through 63 (UUU).

3 Chemistry-Derived Qubit Coordinates

Each codon’s three bases are assigned qubit coordinates (q_x, q_y, q_z) derived from integer-scaled molecular weights:

$$q_x = \text{MW}_{\text{base}_1} \times S, \quad q_y = \text{MW}_{\text{base}_2} \times S, \quad q_z = \text{MW}_{\text{base}_3} \times S \quad (4)$$

where $S = 100$ is the integer scaling factor (multiplying the molecular weight in g/mol by 100 to obtain an integer coordinate) and MW_{base} is the molecular weight of the nucleobase:

- Adenine: 135.13 g/mol
- Cytosine: 111.10 g/mol
- Guanine: 151.13 g/mol

- Uracil: 112.09 g/mol

The qubit coordinates are mapped to Q128.128 fixed-point lattice positions, with surface areas computed from van der Waals (VdW) radii.

4 6D Jordan Routing Channels

Each codon is routed to one of six channels based on amino acid properties:

Table 1: 6D Jordan routing channels.

Channel	Name	Route	Amino Acid Property
E2	Start	AUG	Start codon (Methionine)
E6	Stop	UAA, UAG, UGA	Stop codons
E7	Acidic	Asp (D), Glu (E)	Negative charge
E5	Basic	Lys (K), Arg (R), His (H)	Positive charge
E4	Polar	Ser, Thr, Asn, Gln, Cys, Tyr	Polar uncharged
E0	Nonpolar	Ala, Val, Leu, Ile, Pro, Phe, Trp, Met, Gly	Hydrophobic

5 Smith-Möbius Boundary

The routing system uses two related Möbius transformations:

Smith chart (impedance matching, standard EE):

$$\Gamma_{\text{SC}}(z) = \frac{z - 1}{z + 1} \quad (5)$$

with boundaries $\Gamma_{\text{SC}}(0) = -1$, $\Gamma_{\text{SC}}(1) = 0$, $\Gamma_{\text{SC}}(\infty) = +1$.

Self-inverse checksum ($E=mc^2 \leftrightarrow E=mc^{-2}$ round-trip):

$$\Gamma(z) = \frac{1 - z}{1 + z} \quad (6)$$

Theorem 1 (Self-Inverse Property). $\Gamma(\Gamma(z)) = z$, i.e., the Möbius transformation $\Gamma(z) = (1 - z)/(1 + z)$ is self-inverse.

Proof.

$$\begin{aligned} \Gamma(\Gamma(z)) &= \frac{1 - \frac{1-z}{1+z}}{1 + \frac{1-z}{1+z}} = \frac{\frac{(1+z)-(1-z)}{1+z}}{\frac{(1+z)+(1-z)}{1+z}} \\ &= \frac{(1+z) - (1-z)}{(1+z) + (1-z)} = \frac{2z}{2} = z \quad \square \end{aligned}$$

□

The self-inverse boundaries are:

$$\Gamma(0) = 1, \quad \Gamma(1) = 0, \quad \Gamma(\infty) = -1 \quad (7)$$

Note: $\Gamma_{\text{SC}}(z) = -\Gamma(z)$. The Smith chart formula is NOT self-inverse ($\Gamma_{\text{SC}}(\Gamma_{\text{SC}}(z)) = -1/z$), while the self-inverse form is NOT the standard Smith chart. Both are used for their respective purposes.

6 Cross-Wiring to Octonion Basis

The 64 codons are cross-wired to the octonion basis $\{e_0, e_1, \dots, e_7\}$:

$$\text{Codon}(b_1, b_2, b_3) \rightarrow e_i \quad \text{where } i = \text{routing channel index} \quad (8)$$

The 3-qubit quantum state $|q_x q_y q_z\rangle$ provides a quantum representation:

$$|000\rangle \leftrightarrow e_0, \quad |001\rangle \leftrightarrow e_1, \quad \dots, \quad |111\rangle \leftrightarrow e_7 \quad (9)$$

The 15-layer central row of the 15×15 matrix provides the lattice backbone, with each layer corresponding to an octonion basis element.

7 Quantum Witnesses

The Q# project provides 6 quantum witness operations for codon routing:

Table 2: Q# quantum witness operations for codon routing.

#	Witness	Description
1	Codon encoding	6-qubit encoding of 64 codons
2	Chemistry qubit	3-qubit encoding of molecular weights
3	Channel routing	Phase witnesses for E2/E6/E7/E5/E4/E0
4	Smith-Möbius	Boundary reflection witness
5	Full/stop/acidic	Codon category witnesses
6	Octonion cross-wire	Codon-to-octonion mapping witness

8 Computational Verification

Table 3: Test results for codon routing.

Component	Tests	Status
Zig core (<code>codon.zig</code>)	37	ALL PASS
Q# witnesses	6	ALL PASS
Sidecar f64 validation	16	ALL PASS
Total	59	ALL PASS

The sidecar validation verifies:

- Surface area computation against reference values
- Lattice triangle area (integer cross product + fixed-point sqrt)
- Qubit coordinates for all 64 codons
- Ladder coordinates
- All 64 codon signatures

9 AUG and GCA Outliers

Two codons exhibit special behavior:

- **AUG (Methionine/Start)**: Detected by the ChemistrySignatureBuilder as the start codon. AUG is the only codon encoding Methionine and serves as the initiation signal for translation.
- **GCA (Alanine)**: Detected by the PlaceholderSignatureBuilder as a special case. GCA is one of four codons encoding Alanine, but its chemistry signature deviates from the expected pattern.

10 Scientific Scope

The codon routing system is a **framework-internal heuristic construction**. Specifically:

- The mapping from codons to Jordan algebra channels is **not** derived from biological first principles.
- The chemistry-derived qubit coordinates are **heuristic**—they use molecular weights as a proxy for quantum information content.
- The cross-wiring to octonions is a **construction**—it is built to match, not derived.
- The Smith-Möbius boundary is **mathematically correct** (the self-inverse property is proven), but its biological relevance is **unverified**.
- The $Q\#$ witnesses verify the **internal consistency** of the encoding, not its biological validity.

This work should be understood as a computational exploration of algebraic structures in biological information, not as a claim about the biological mechanism of translation or protein folding.

11 Conclusion

We have presented a computational mapping from the 64-codon genetic code to 6D Jordan algebra routing channels. The mapping uses chemistry-derived qubit coordinates, six routing channels based on amino acid properties, and cross-wiring to the octonion basis. The system is verified with 59 tests across Zig, $Q\#$, and f64 sidecar implementations. The mapping is a heuristic construction—computationally consistent but not biologically established. Future work could explore whether the algebraic structure of the genetic code has biological significance beyond the framework’s internal construction.

Acknowledgments

Computational assistance was provided by Devin AI, Chat-GPT, Gemini, GLM-2.5, and Kimi AI. These AI tools assisted with code generation and documentation but did not author the mathematical content, which is the sole responsibility of the corresponding author.

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